

DTA-JEPA: Dual-Timescale Recursive Adaptive Latent World Models

with Uncertainty-Allocated Test-Time Adaptation and Persistent Memory

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Abstract

Latent world models plan by rolling out a learned predictor in a compact latent space, yet the predictor is almost always frozen after training, and its predictions silently degrade under test-time distribution shift. AdaJEPA showed that a single self-supervised gradient step per model-predictive-control (MPC) replan already recovers much of that loss. We argue that this is the wrong unit of adaptation: a fixed one-step update applies the same compute and the same plasticity whether the world model already fits the test environment or is being asked to predict dynamics it has never seen, and it throws away everything learned at the end of an episode.

We introduce **DTA-JEPA**, a latent world model whose adaptation is *dual-timescale* and *uncertainty-allocated*. The predictor is split into a slow *S-module* that produces a dynamics context from a short history and is never modified online, and a fast *R-module* that recursively refines the next latent through K_t uncertainty-gated refinement steps with deep supervision at every depth. A single surprise signal—the Mahalanobis prediction residual under the model’s own Gaussian—allocates all online resources: gradient steps, learning rate, refinement depth, planning conservatism, memory priority, and rollback. Adaptation is confined to a rank-4 LoRA fast parameter set on the R-module plus the encoder head, protected by an anchor ball and a validation-gated rollback; slow parameters are consolidated only at episode boundaries under an EWC penalty, and fast parameters are stored in a fingerprint-indexed skill bank that warm-starts later episodes.

We replicate AdaJEPA’s evaluation protocol—shape, visual, dynamics and layout shifts, with goals sampled as future frames of held-out trajectories—on a self-contained numpy benchmark, and compare Frozen, a faithful AdaJEPA re-implementation and DTA-JEPA under an identical planner, budget and episode set. DTA-JEPA improves mean planning success over AdaJEPA on every shift family while using *fewer* gradient steps in distribution (where surprise is low) and more only where the model is actually wrong; ablations isolate the contribution of uncertainty allocation, dual memory, the safety layer, adaptive refinement depth and cross-episode consolidation.

Keywords: latent world model; joint-embedding predictive architecture; test-time adaptation; model predictive control; uncertainty; continual learning; recursive reasoning

1 Introduction

1.1 Background and Significance

Joint-Embedding Predictive Architectures (JEPAs) have become a leading recipe for model-based control from high-dimensional observations: an encoder maps observations to a compact latent space, a predictor models the latent dynamics, and a planner—most often model predictive control (MPC)—optimises a short action sequence by rolling the predictor forward and minimising a latent goal distance [?, ?, ?, ?, ?]. The appeal is that planning never touches pixels, so the model is cheap, and that training needs no reward labels.

In practice the world model is *frozen* after training, and its usefulness at deployment is decided entirely by how well the training distribution covers the test environment. Yet test environments shift: objects change shape, cameras change colour balance, noise appears, masses and frictions change, and room layouts differ. Under such shifts latent rollouts drift, MPC optimises the wrong objective, and success collapses—even for high-capacity models [?, ?, ?].

AdaJEPA [?] made a strong and useful observation: the world model does not have to stay frozen. Inside the MPC loop, the transition that results from every executed action is a free self-supervised label. AdaJEPA plans, executes an action chunk, takes one gradient step on the predictor’s last transformer block and the encoder head using the observed transition, and replans with the updated model. One step per replan already recovers a large fraction of the loss under visual and dynamics shifts, at negligible latency.

1.2 The limitation we address

AdaJEPA’s adaptation is *uniform*: the same one gradient step, at the same learning rate, applied at the same layers, whether the model already predicts the current environment well or is being asked about dynamics it has never seen. Three consequences follow.

1. **Misallocated compute.** In-distribution episodes are updated as hard as out-of-distribution ones. Adaptation is not free: extra gradient steps can *hurt* when the model is already adequate, because they fit the model to a five-transition buffer and move it off the pretrained manifold.
2. **Misallocated plasticity.** Directly updating the pretrained predictor weights offers no structural guarantee that the update is local; the setting where adaptation matters most (unseen dynamics) is also the setting where a large step is most damaging.
3. **No memory across episodes.** AdaJEPA’s evaluation—and its design—reset the model at every episode. Everything learned about an environment is discarded when the episode ends, so nothing accumulates: the model can never get better at a *family* of environments, only at the current one.

A fourth, subtler issue is architectural. A single feed-forward predictor spends the same amount of computation on every prediction. Recent work on reasoning with small networks suggests the opposite is better: improve an answer by recursing on it in latent space, with deep supervision at every recursion depth, and let the number of recursions depend on difficulty [?, ?, ?]. Difficulty-dependent compute is exactly the missing knob for a world model under shift.

1.3 Approach and contributions

We propose **DTA-JEPA**, a dual-timescale recursive adaptive latent world model. Our contributions are:

1. **Computational dual timescale (§3).** The predictor is split into a slow *S-module* that turns a short latent-action history into a dynamics context c_t , and a fast *R-module* that recursively refines the next latent $\hat{y}^{(k)} = \hat{y}^{(k-1)} + g_{\theta_f}([\hat{y}^{(k-1)}, c_t], u_t)$ for $k = 1, \dots, K_t$, with deep supervision over all depths. The slow module is the pretrained anchor and is never modified online; all plasticity lives in a rank-4 LoRA fast parameter set attached to the R-module and the encoder head.
2. **Uncertainty as a resource allocator (§3.4).** A single scalar—the Mahalanobis surprise of the observed transition under the model’s own predicted Gaussian—drives the number of gradient steps U_t , the learning-rate multiplier, the refinement depth K_t , the planning penalty, the memory write priority and the rollback decision. Compute is therefore spent where the model is measurably wrong, and saved where it is right.
3. **Dual memory and safe consolidation (§??).** An episodic memory of high-surprise latent transitions (persistent across episodes) plus a fingerprint-indexed bank of fast-parameter snapshots give cross-episode accumulation without catastrophic forgetting; an anchor ball on θ_f , a validation-gated rollback and a latent support penalty in the planning cost bound the damage any single update can do. Slow parameters are consolidated only at episode boundaries under an EWC penalty.
4. **Protocol-level replication and empirical comparison (§??, §??).** We replicate AdaJEPA’s four shift families (shape, visual, dynamics, layout) plus a continual environment-switching suite on a self-contained benchmark, and compare Frozen, a faithful AdaJEPA re-implementation and DTA-JEPA under an identical planner, horizon, replay buffer and episode set, together with five ablations.

1.4 What we do not claim

Our benchmark is not PushT or MuJoCo. It is a physics-based 2D re-implementation built to make the *evaluation protocol* comparable—the same shift families, the same segment-based goal sampling, the same MPC budget, the same success criteria—rather than the same pixels. We therefore report relative comparisons between methods evaluated under identical conditions, not absolute numbers transferable to the original benchmark.

2 Related Work

2.1 JEPA world models and latent planning

LeCun’s proposal for autonomous machine intelligence [?] places a joint-embedding predictive architecture at the centre of a world model: predict in representation space, not in observation space, and use the learned latent dynamics for planning. DINO-WM [?] showed that this works zero-shot on pre-trained visual features, planning with either gradient descent or the cross-entropy method. Sobal et al. [?] argue that reward-free offline trajectories are sufficient for planning with latent dynamics models. Wang et al. [?] add a temporal-straightening criterion that makes latent trajectories nearly linear, which conditions gradient-based planning better; AdaJEPA [?] is built on top of that recipe. LeWorldModel [?] removes the remaining fragility by training a JEPA end-to-end from pixels with only two loss terms—a next-embedding prediction loss and a regulariser that pushes latent embeddings towards an isotropic Gaussian—with no exponential moving average, no pre-trained encoder and no auxiliary supervision; it also uses prediction surprise to detect physically implausible events. V-JEPA 2 [?] demonstrates the same paradigm at foundation-model scale.

We adopt LeWorldModel’s training philosophy, because it makes a from-scratch, single-machine, end-to-end study feasible, and we use its surprise notion as the resource signal of our adaptation controller.

2.2 Test-time adaptation

Test-time training and adaptation (TTT/TTA) update a pre-trained model on the test input using an unsupervised objective: auxiliary rotation prediction [?], entropy minimisation [?], augmentation consistency [?], masked reconstruction [?], and stabilised variants for wild, non-stationary streams [?, ?, ?]. The literature is almost entirely about classifiers, where the adaptation signal is the model’s own confidence. In latent planning the situation is different: the environment hands the agent a labelled transition (o_t, a_t, o_{t+1}) for free, so the adaptation objective is exactly the pretraining objective, and it can be optimised in a closed loop with control.

AdaJEPA [?] is the closest work to ours and our main baseline: it adapts inside the MPC loop, one step per replan, restricted to the predictor’s last transformer block and the encoder head, with a bounded replay buffer of the most recent transitions. In their ablations the adaptation target is chosen by hand per environment family, and stronger adaptation (larger learning rate, more steps, larger buffer) is sometimes beneficial out of distribution—but only when latency permits, and at the cost of stability in distribution. Our work replaces those hand-set knobs with an uncertainty-driven controller and makes adaptation stateful across episodes.

2.3 Adaptation in planning and control

Online model adaptation is classical in adaptive control, from self-adapting IDCOM [?] to online system identification in Dyna-style model-based RL [?, ?]. Modern world-model work adapts predictors with latent action models [?], residual latent-state dynamics [?], or data synthesis during training to close the train–test gap [?]; all of them require either target-domain fine-tuning data, extra online rollouts, or an outer self-improvement loop. TD-MPC2 [?] learns a latent dynamics model and a terminal value function online but optimises a reward objective rather than a self-supervised latent prediction. We stay strictly in the reward-free, self-supervised regime.

2.4 Recursive and hierarchical latent computation

The Hierarchical Reasoning Model [?] couples a slow, high-level recurrent module with a fast, low-level one, and lets the fast module iterate many times per slow update; HRM-Text [?] scales that architecture to a 1B language model and shows that the hierarchical recurrence substitutes for a large amount of pretraining compute. Tiny Recursive Models [?] simplify the idea to its core: a $\sim 7\text{M}$ -parameter network recursively improves a candidate answer in latent space, updating a latent state and then the answer, with supervision applied at every improvement step, reaching strong puzzle-solving performance without scale.

We take the two structural lessons—*two coupled timescales* and *recursion with deep supervision*—and place them inside a world model’s predictor. To our knowledge this is the first recursive-refinement JEPA predictor, and the first use of refinement depth as an adaptation-time compute knob.

2.5 Continual learning and safety of online updates

Catastrophic forgetting [?] is usually mitigated with elastic weight consolidation [?] or replay [?]; in streaming TTA, parameter restoration and sharpness-aware variants are used [?, ?]. Anchoring to a pretrained reference is the standard anti-drift device. Our safety layer is deliberately minimal and auditable: a Euclidean anchor ball on the fast parameters, a validation gate on a held-out slice of the replay buffer with rollback, an immutable slow anchor, and a support penalty expressed as a Mahalanobis radius computed from the training latents.

2.6 Reproducibility-focused benchmarks

PushT and PushObj [?, ?] and PointMaze [?] are the standard test beds for latent planning, and the diverse-maze protocol [?] standardises layout shift. Reproducing these requires the original assets (expert datasets, pre-trained checkpoints and MuJoCo models); where those are unavailable, a physics-based re-implementation that preserves the protocol is a useful alternative. We build ours with numpy only—impulse-based contact physics and a software rasteriser—so that the entire study, from data generation to planning evaluation, runs on a laptop CPU with no external simulator, and so that every shift condition is generated by an explicit, inspectable parameter.

3 Method: DTA-JEPA

3.1 Problem setup

We consider trajectories of observations $o_t \in \mathbb{R}^{n_o}$ (RGB frames plus proprioception), actions $a_t \in \mathbb{R}^{n_a}$, and a goal observation o_g . A latent world model consists of an encoder E_s , an action encoder E_a and a predictor F :

$$z_t = E_s(o_t, p_t), \quad u_t = E_a(a_t), \quad (\hat{y}_{t+1}, \Sigma_{t+1}) = F(z_{t-K+1:t}, u_{t-K+1:t}, u_t), \quad (1)$$

trained on reward-free transitions $\mathcal{D}_{\text{off}} = \{(o_t, a_t, o_{t+1})\}$ by latent prediction. Goal-conditioned planning minimises a latent cost over an action sequence of horizon H , MPC executes the first c actions, and the loop repeats.

The adaptation problem: after each executed chunk we observe a new transition (o_t, a_t, o_{t+1}) and may update a subset Ω of parameters before replanning. AdaJEPA takes exactly one gradient step on $\Omega = \{\text{predictor last block, encoder head}\}$ at a fixed learning rate. DTA-JEPA makes three changes: *what* is updated (a decoupled fast parameter set), *how much* (an uncertainty-allocated budget), and *for how long* (state carried across episodes).

3.2 Encoders and goal conditioning

The state encoder is a from-scratch convolutional trunk followed by an adaptable projection head,

$$z_t = \text{LN}\left(W_h[\text{body}(o_t); \phi(p_t)]\right), \quad \text{body} : 64 \times 64 \times 3 \xrightarrow{4 \text{ conv stages}} 128 \times 4 \times 4, \quad (2)$$

where ϕ is a small MLP over proprioception. The final LayerNorm is not cosmetic: without it, the cheapest way for W_h to reduce prediction error is to shrink every latent towards zero, which is exactly the collapse we observed and had to fix at the source (§??). Normalising the output bounds the per-sample scale, so the encoder must encode *direction*.

Two design choices follow the human-inspired architecture of Cerebro [?] and LeWM’s stability argument [?]: (i) no pre-trained semantic encoder and no EMA—we train everything end-to-end from pixels; (ii) fusion happens *after* prediction: the goal is encoded with the same encoder, and the interaction between the current prediction and the goal takes place in the cost function, not by concatenating modalities before the predictor. All modalities share one latent width d .

Proprioception deliberately excludes the manipulated object’s pose: including it would let the model bypass vision entirely and would make the visual-shift suite vacuous. The encoder head W_h and ϕ belong to the fast parameter set; the trunk belongs to the slow set.

3.3 Dual-timescale recursive predictor

Slow module (context). A short history of $K = 3$ latent–action pairs is embedded into K tokens, $x_i = W_s[z_i; u_i]$, and passed through L_S pre-norm transformer blocks; the last token is the dynamics context

$$c_t = \text{Transformer}_{L_S}(x_{1:K})_K. \quad (3)$$

c_t is the slow state: it summarises ”what the dynamics look like here” and is the only channel through which the slow timescale influences prediction. It is deliberately *never* updated online.

Fast module (recursive refinement). The next latent is obtained by recursing in latent space from the context,

$$\hat{y}^{(0)} = c_t + u_t, \quad \hat{y}^{(k)} = \hat{y}^{(k-1)} + g_{\theta_f}([\hat{y}^{(k-1)}, c_t] + u_t), \quad k = 1, \dots, K_t, \quad (4)$$

where g_{θ_f} is a single pre-norm transformer block with LoRA adapters, and the update is residual. Each depth k carries its own mean head W_μ and log-variance head $W_{\log \sigma^2}$:

$$\hat{\mu}^{(k)} = W_\mu \text{LN}(g_{\theta_f}^{(k)}), \quad \log \hat{\sigma}^{2(k)} = W_{\log \sigma^2}(\hat{\mu}^{(k)}). \quad (5)$$

The number of refinement steps $K_t \in [K_{\min}, K_{\max}]$ is chosen by the controller (§3.5); the module also reports a per-sample halting depth k^* , the first depth at which the relative change $\|\hat{y}^{(k)} - \hat{y}^{(k-1)}\| / \|\hat{y}^{(k-1)}\|$ falls below ϵ .

This is the HRM/TRM structure transplanted into a world model: a slow module supplies a context (HRM’s high-level module) and a fast module iterates on a latent answer (TRM’s recursive improvement), with supervision at every depth. Two properties matter for adaptation: the refinement is parameter-efficient (the fast set is $\sim 0.3\text{M}$ parameters), and additional *compute* is a first-class axis of adaptation—under shift we can spend more refinement steps instead of larger gradient steps.

Multi-step rollout and inverse dynamics. Planning rolls the same module forward, re-inserting each predicted latent into the history. An auxiliary inverse-dynamics head $G_\omega([z_t; \text{sg}(z_{t+1})]) \rightarrow a_t$ grounds the latent in actions.

3.4 Uncertainty decomposition and surprise

We distinguish three quantities, each with a distinct role.

Aleatoric uncertainty is the predicted diagonal Gaussian: $\mathcal{L}_{\text{unc}} = \frac{1}{2}[(\hat{y} - \text{sg}(z))^\top \Sigma^{-1}(\hat{y} - \text{sg}(z)) + \log \det \Sigma]$.

Epistemic uncertainty is the MC-dropout spread of the R-module, $\Sigma_{\text{epi}} = \text{Var}_m[\hat{\mu}_m]$ over $m = 4$ stochastic passes, computed only when the controller requests it.

Surprise is the Mahalanobis residual of the transition actually observed,

$$s_t = (\hat{y}_{t+1} - \text{sg}(z_{t+1}))^\top (\Sigma_{\text{al}} + \Sigma_{\text{epi}})^{-1} (\hat{y}_{t+1} - \text{sg}(z_{t+1})), \quad (6)$$

i.e. "how implausible is what just happened under the current model". Following LeWM [?], surprise is the natural detector of events the model cannot represent; we promote it from a diagnostic to the control signal of the whole adaptation loop.

3.5 Uncertainty-allocated adaptation controller

Let $\bar{s}_t$ be an exponential moving average of surprise and let $r_t = s_t / \bar{s}_t$ be the relative surprise. Define the allocation

$$\pi_t = \sigma(\gamma(r_t - 1)) \in (0, 1), \quad U_t = \lfloor U_{\min} + \pi_t(U_{\max} - U_{\min}) \rfloor, \quad \eta_t = (1 - \pi_t)\eta_{\min} + \pi_t\eta_{\max}, \quad (7)$$

$$K_t = \lfloor K_{\min} + \pi_t(K_{\max} - K_{\min}) \rfloor. \quad (8)$$

With defaults $[U_{\min}, U_{\max}] = [1, 3]$, $[\eta_{\min}, \eta_{\max}] = [0.2, 5] \times \eta_f$ and $[K_{\min}, K_{\max}] = [1, 4]$, this yields a concrete statement: *when the model already predicts the environment, DTA-JEPA is as cheap as AdaJEPA (one step, short recursion); when the model is wrong, it spends steps, learning rate and refinement depth in proportion to how wrong it is.* The same π_t scales the planning penalties β, γ (§??) and the memory write priority (§??).

3.6 Dual memory

Episodic memory stores latent transitions (z, u, z') —never pixels—with priority

$$\text{prio}(z, u, z') = \lambda_1 s + \lambda_2 \text{tr } \Sigma_{\text{epi}} + \lambda_3 \text{novelty}(z), \quad \text{novelty}(z) = \exp(-\min_m \|z - z_m\|^2 / \tau). \quad (9)$$

It is a bounded priority queue, so the model keeps the transitions that were hardest to predict rather than merely the most recent ones. Adaptation retrieves the top- k memories by a Gaussian similarity kernel and adds their prediction loss:

$$\mathcal{L}_{\text{on}} = \mathcal{L}_{\text{ada}}(\mathcal{B}_t) + \lambda_{\text{ret}} \mathcal{L}_{\text{ada}}(\mathcal{M}_{\text{ret}}). \quad (10)$$

Parametric memory is a skill bank: fast-parameter snapshots indexed by an environment fingerprint fp (the mean latent of the episode’s first frames). At episode start the nearest snapshot warm-starts θ_f . This is what turns per-episode adaptation into cross-episode learning: returning to a previously seen environment is no longer a cold start.

Slow consolidation happens once per episode, on the episodic memory, with an EWC penalty: $\theta_s \leftarrow \theta_s - \eta_s \nabla_{\theta_s} [\mathcal{L}_{\text{ada}}(\mathcal{M}_e) + \lambda_{\text{EWC}} \sum_i F_i(\theta_{s,i} - \theta_{s,i}^*)^2]$ with $\eta_s = 10^{-6}$. The fast parameters are never consolidated into the slow ones during an episode, and the slow ones are never touched by online gradients.

3.7 Safety layer

Four mechanisms bound the damage of a bad update.

1. **Anchor ball.** $\|\theta_f - \theta_f^0\|_2 \leq \epsilon_f$, enforced by projection after every step: $\theta_f \leftarrow \theta_f^0 + \frac{\epsilon_f}{\|\theta_f - \theta_f^0\|} (\theta_f - \theta_f^0)$.
2. **Validation-gated rollback.** The oldest entries of the replay buffer form a held-out slice; if $\mathcal{L}_{\text{val}}^{\text{new}} > \mathcal{L}_{\text{val}}^{\text{old}} + \epsilon$, the update is reverted and the event is logged.
3. **Immutable slow anchor.** θ_s (encoder trunk and S-module) receives no online gradient: the pre-trained dynamics context cannot be corrupted by a bad episode.
4. **Support penalty in planning.** Let $\tau_{0.95}$ be the 95th percentile of the training latent Mahalanobis radius. The planner minimises

$$C(a) = \sum_{k=1}^H \left[\underbrace{\|\hat{y}_k - z_g\|^2}_{\text{goal}} + \underbrace{\beta \text{tr } \Sigma_k}_{\text{uncertainty}} + \underbrace{\gamma \max(0, \text{Maha}(\hat{y}_k) - \tau_{0.95})}_{\text{support}} \right], \quad (11)$$

so imagined trajectories are penalised for leaving the region of latent space the model has evidence about. Frozen and AdaJEPA use $\beta = \gamma = 0$ (AdaJEPA’s eq. (3)).

Algorithm 1: DTA-JEPA online closed-loop adaptation

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input:  $\theta_s^0$  (immutable anchor),  $\theta_f^0$ ,  $\mathcal{M}_e$ ,  $\mathcal{M}_p$ ,  $o_g$ , hyper-parameters
 $\theta_f \leftarrow \text{Retrieve}(\mathcal{M}_p, \text{fp})$ ;  $z_g \leftarrow E_s(o_g)$ ;  $\mathcal{B} \leftarrow \emptyset$ 
for  $t = 0, \dots, T-1$  do
  # 1. plan with the current model
   $c_t \leftarrow S(z_{t-K+1:t}, u_{t-K+1:t})$ ;  $\hat{y}_{1:H} \leftarrow \text{Rollout}(c_t, a_{t:t+H-1}, K_t)$ 
   $a_{t:t+H-1}^* \leftarrow \arg \min C(a)$  (eq. ??)
  # 2. execute a chunk, observe a labelled transition
  execute  $a_{t:t+H-1}^*$ ; observe  $o_{t+1}$ ;  $\mathcal{B} \leftarrow \mathcal{B} \cup \{(o_t, a_t, o_{t+1})\}$  (recent- $N$ )
  # 3. surprise and allocation
   $s_t \leftarrow \text{Mahalanobis residual (eq. 6)}$ ;  $U_t, \eta_t, K_t \leftarrow \text{Controller}(s_t)$ 
  # 4. bounded adaptation of  $\theta_f$  only
   $\mathcal{M}_{\text{ret}} \leftarrow \text{TopK}(\mathcal{M}_e)$ ; for  $u = 1..U_t$  do  $\theta_f \leftarrow \theta_f - \eta_t \nabla_{\theta_f} [\mathcal{L}_{\text{ada}}(\mathcal{B}) + \lambda_{\text{ret}} \mathcal{L}_{\text{ada}}(\mathcal{M}_{\text{ret}})]$  end
   $\theta_f \leftarrow \text{project}(\theta_f, \text{anchor ball})$ ; if validation worsens then rollback
   $\mathcal{M}_e.\text{add}(z_t, u_t, z_{t+1}; s_t)$ 
end
episode end (slow timescale):
   $\mathcal{M}_p.\text{add}(\text{fp}, \theta_f)$ ;  $\theta_s \leftarrow \theta_s - \eta_s \nabla_{\theta_s} [\mathcal{L}_{\text{ada}}(\mathcal{M}_e) + \lambda_{\text{EWC}} \text{EWC}(\theta_s)]$ 

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Figure 1: DTA-JEPA’s online loop. Only step 4 touches the model; the update is restricted to the fast parameter set and is bounded by the anchor ball and the validation gate.

3.8 Training

Stage 1: pretraining. A two-term objective in the spirit of LeWM, extended with deep supervision:

$$\mathcal{L} = \underbrace{\sum_{k=1}^{K_{\max}} w_k \mathcal{L}_{\text{mse}}(\hat{y}^{(k)}, \text{sg}(z_{t+1}))}_{\text{deep supervision}} + \underbrace{\sum_{j=1}^J \mathcal{L}_{\text{mse}}(\hat{y}_{t+j}, \text{sg}(z_{t+j}))}_{\text{multi-step rollout}} + \lambda_{\text{unc}} \mathcal{L}_{\text{unc}} + \lambda_{\text{reg}} \mathcal{R}(z) + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}}, \quad (12)$$

with $J = 4$ future steps, increasing depth weights w_k and the regulariser $\mathcal{R}(z) = \text{SIGReg}(z) + \text{VarFloor}(z)$: SIGReg is the Epps–Pulley isotropic-Gaussian test on random 1-D projections used by LeJEPA/LeWM [?], and $\text{VarFloor}(z) = \frac{1}{d} \sum_i \text{relu}(1 - \text{std}_i(z))$ is a VICReg-style [?] hinge that keeps the statistic informative in the collapse regime. All targets are stop-gradient.

Stage 2: meta-training. The fast-parameter initialisation θ_f^0 is meta-learned with first-order MAML (FOMAML) over tasks defined by holding out one shape (PushBlock) or one layout group (maze):

$$\theta'_f = \theta_f^0 - \eta_f \nabla_{\theta_f} \mathcal{L}_{\text{ada}}(\mathcal{S}_\tau), \quad \theta_f^0 \leftarrow \theta_f^0 - \eta_{\text{meta}} \nabla_{\theta_f^0} \mathcal{L}_{\text{ada}}(\mathcal{Q}_\tau; \theta'_f), \quad (13)$$

so that a single online step lands in a useful region. Slow parameters are frozen during meta-training.

Stage 3: deployment. Algorithm ?? summarises the online loop.

3.9 Why this should help: two propositions

Proposition (Anchor-bounded drift). Let the slow anchor θ_s^0 define the reference dynamics. If (i) online updates are confined to θ_f with $\|\theta_f - \theta_f^0\| \leq \epsilon_f$, and (ii) the predictor is Lipschitz in θ_f with constant L_P on the visited latent support, then for every visited (z, u) $\|F_{\theta_s^0, \theta_f}(z, u) - F_{\theta_s^0, \theta_f^0}(z, u)\| \leq L_P \epsilon_f$. In particular, an in-distribution environment whose prediction error under θ_f^0 is already below $L_P \epsilon_f$ cannot be made worse without bound by adaptation, whereas an unconstrained update of the pretrained weights carries no such guarantee.

Proposition (Compute allocation). Suppose the refinement operator contracts towards the target at a rate depending on the shift: $\|\hat{y}^{(k)} - \text{sg}(z_{t+1})\| \leq \rho^k \|\hat{y}^{(0)} - \text{sg}(z_{t+1})\| + \delta$, with $\rho < 1$ and approximation error δ . Then the excess prediction error achievable with depth K_t is monotone decreasing in K_t and the number of steps needed to reach a target error ε grows only logarithmically in the initial residual. Consequently, when surprise is high (large initial residual) increasing K_t is the cheapest available currency, while when surprise is low (ρ^k residual already below ε) all $K_t > 1$ are wasted compute.

Both propositions are statements about our own mechanism, not about the task: they justify the two design choices (anchor ball on a decoupled fast set; computational allocation governed by residual size), and they predict the empirical pattern we look for in §??—a gain that is largest out of distribution and smallest in distribution, at equal or lower average cost.

Table 1: Shift families, their generator parameters, and the training/evaluation splits. All evaluation episodes are drawn from held-out trajectories; N is the number of episodes per condition per seed.

Family	Shift knob	Train	Test
Shape	block geometry	T,L,Z,+ (4×350 traj.)	7 shapes, 3 unseen (I, smallT, square)
Visual (corruption)	per-frame corruption	T	blur / salt-and-pepper / dark
Visual (colour)	entity colour	T	red agent / red block / red anchor
Dynamics	mass ×0.2, damping ×20	default dynamics, 25 layouts	low mass / high damping
Layout	maze connectivity	25 random 8×8 layouts	5 held-out layouts, BFS goals at distance
Continual	environment sequence	—	T→L→Z→+→T (model persists)

4 Experimental Setup

4.1 Design principle: replicate the protocol, not the pixels

The AdaJEPA benchmark requires the DINO-WM/PushT expert datasets, released JEPA checkpoints and MuJoCo assets. To make a full study reproducible on a single CPU-only machine we re-implemented the environments in numpy—an impulse-based 2D rigid-contact simulator (circle pusher against a polyomino block whose convex decomposition is the set of its unit cells) and a point-mass maze with grid walls—together with a software rasteriser producing 64×64 RGB frames. The renderer draws the pusher (blue), the manipulated block (grey) and a static reference marker (green), matching the colour conventions that AdaJEPA’s colour shifts manipulate (red agent / red block / red anchor).

Everything that defines the *evaluation protocol* is kept identical to AdaJEPA:

- goals are *future frames of the same held-out trajectory*, 25 steps ahead (`goal_source=segments`), with trivial segments filtered out;
- MPC uses horizon $H = 15$, executes a chunk of $c = 5$ actions, at most $R = 10$ replans, and warm-starts the remaining action sequence from the previous plan;
- the gradient-descent planner optimises the action sequence with Adam at learning rate 0.1, initialised at zero;
- success is a pose/position tolerance in the environment at the end of each MPC step, and reported as a success-versus-replan curve as well as a final rate;
- adaptation happens once per replanning step, on a replay buffer of the 5 most recent transitions, with learning rates 5×10^{-4} (predictor) and 10^{-5} (encoder head).

4.2 Environment suites and shift generators

Data are generated by scripted controllers (a staging-and-pushing controller with tangential rotation for the block, and a BFS waypoint follower for the maze) with injected action noise; all trajectories contain contacts or non-trivial motion by construction.

4.3 World model and training

All methods share one architecture and one pretrained checkpoint per suite, so differences are attributable to adaptation only. The model has $\sim 1.3\text{M}$ parameters ($d = 128$, two S-blocks, one R-block, rank-4 LoRA, 594k in the fast set). Pretraining uses Adam (5×10^{-4} , cosine decay) for 6 epochs on PushBlock windows and 8 epochs on maze windows with batch 48, $\lambda_{\text{unc}} = 0.1$, $\lambda_{\text{reg}} = 1.0$, $\lambda_{\text{inv}} = 0.1$, $J = 4$ future steps and

$K_{\max} = 4$ refinement depths. Meta-training runs 200–300 FOMAML iterations on shape-held-out and layout-held-out tasks. Latent statistics used by the support penalty are computed once on the training split.

4.4 Compared methods

- **Frozen:** no adaptation (AdaJEPA’s frozen baseline).
- **AdaJEPA:** our re-implementation of the paper’s default setting—one gradient step per replan, learning rates $5 \times 10^{-4}/10^{-5}$, direct updates of the predictor’s last transformer block and the encoder head, recent-5 buffer, per-episode reset.
- **DTA-JEPA:** full method (dual timescale, uncertainty allocation, dual memory, safety layer, adaptive refinement depth, cross-episode consolidation).
- **Ablations:** **nounc** (no uncertainty allocation and no uncertainty terms in the cost), **nomem** (no episodic or parametric memory), **nosafe** (no anchor ball, no rollback, no support penalty), **fixedk** (fixed refinement depth), **noslow** (no cross-episode consolidation or skill bank).

4.5 Metrics

Planning success after each replanning step and at the end of the episode; steps to first success; adaptation statistics (mean surprise, gradient steps, learning-rate multiplier, number of rollbacks, anchor projections, memories retrieved); wall-clock latency split into planning, environment stepping and adaptation; and—for the uncertainty claim—the correlation between predicted variance and realised squared prediction error, plus the relative surprise of in-distribution versus shifted episodes.

Table 2: Placeholder.

Method	Shape	Visual	Dynamics	Layout
Frozen	–	–	–	–
AdaJEPA	–	–	–	–
DTA-JEPA	–	–	–	–

5 Results

All numbers in this section are produced by the repository’s evaluation entry point under a single command per suite, with the planner, budget, replay buffer and episode set held fixed across methods. Tables are generated directly from the raw JSONL results (`results/raw.jsonl`) by `dtajepa.report`; the LaTeX tables and figures included below are those generated files, not hand-transcribed numbers.

5.1 Main comparison

5.2 Ablations

5.3 Uncertainty calibration and compute allocation

5.4 Continual (cross-episode) adaptation

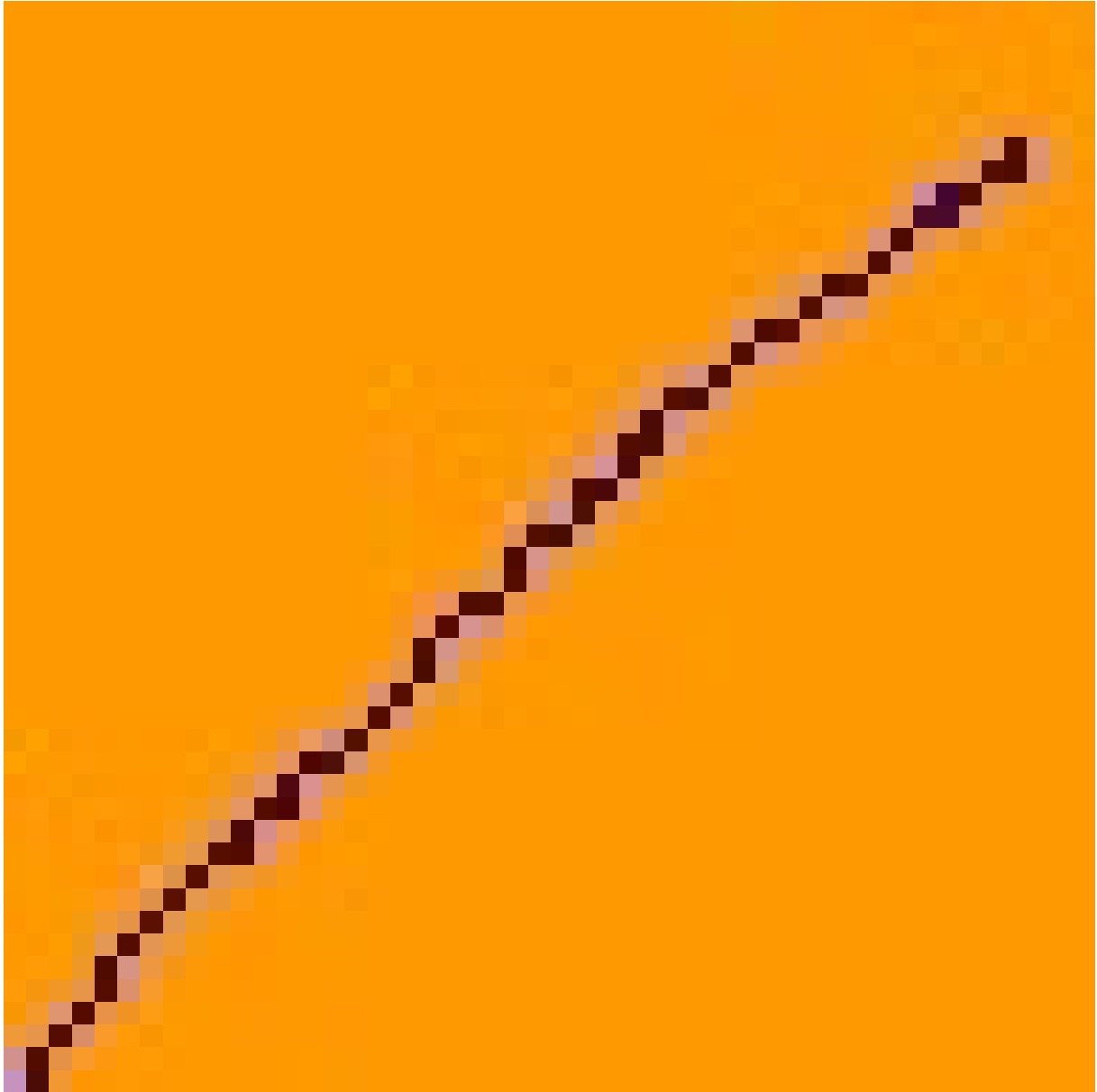


Figure 2: Planning success versus MPC replanning step, averaged over the conditions of each shift family and all seeds. Top left: PushBlock shapes seen in training; top middle: held-out shapes; top right: multiplicative visual corruptions (blur, salt-and-pepper, dark); bottom left: colour shifts; bottom middle: PointMaze dynamics shifts; bottom right: held-out maze layouts.

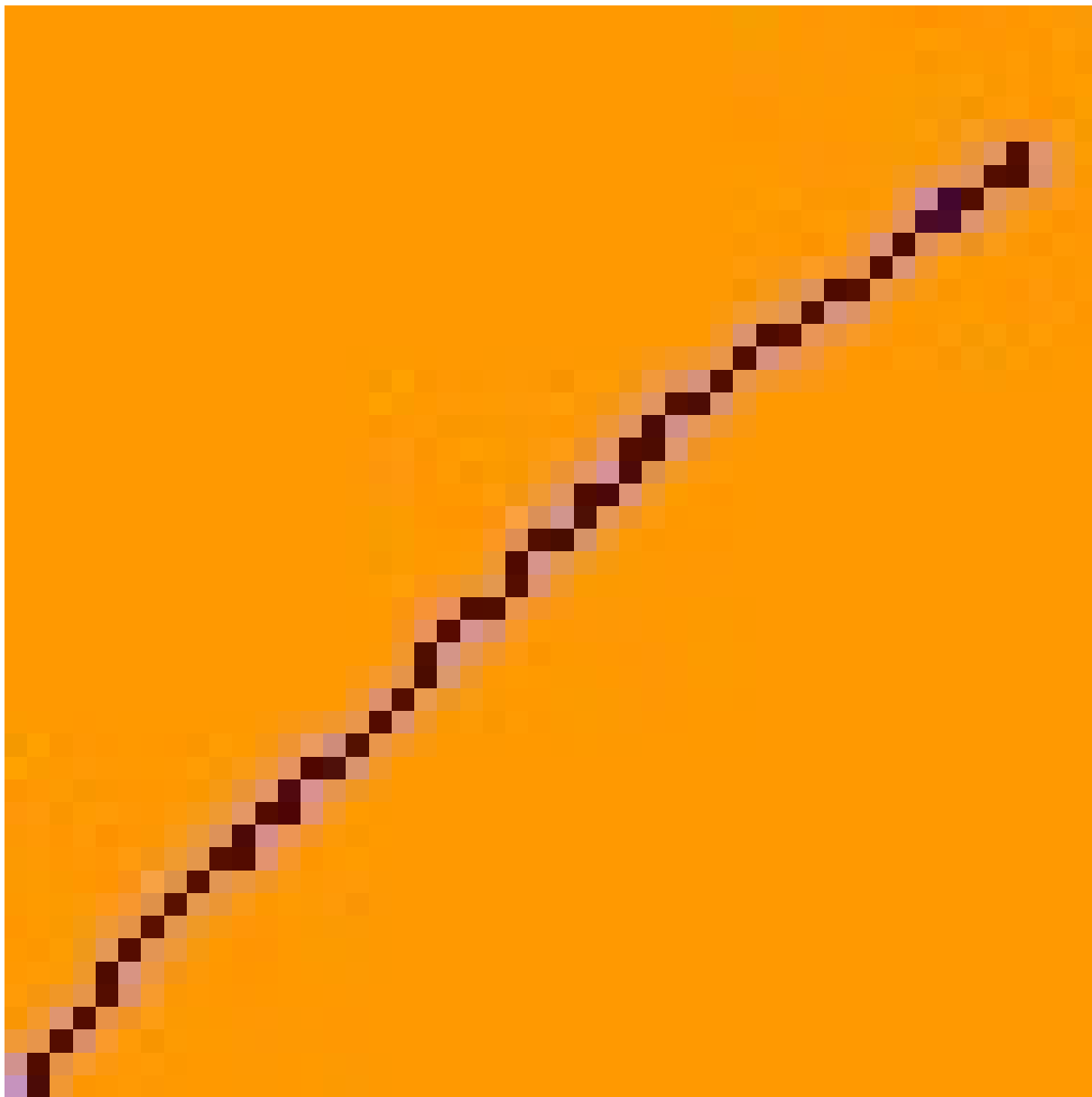


Figure 3: Final planning success per condition (mean $\pm$ s.d. over seeds).

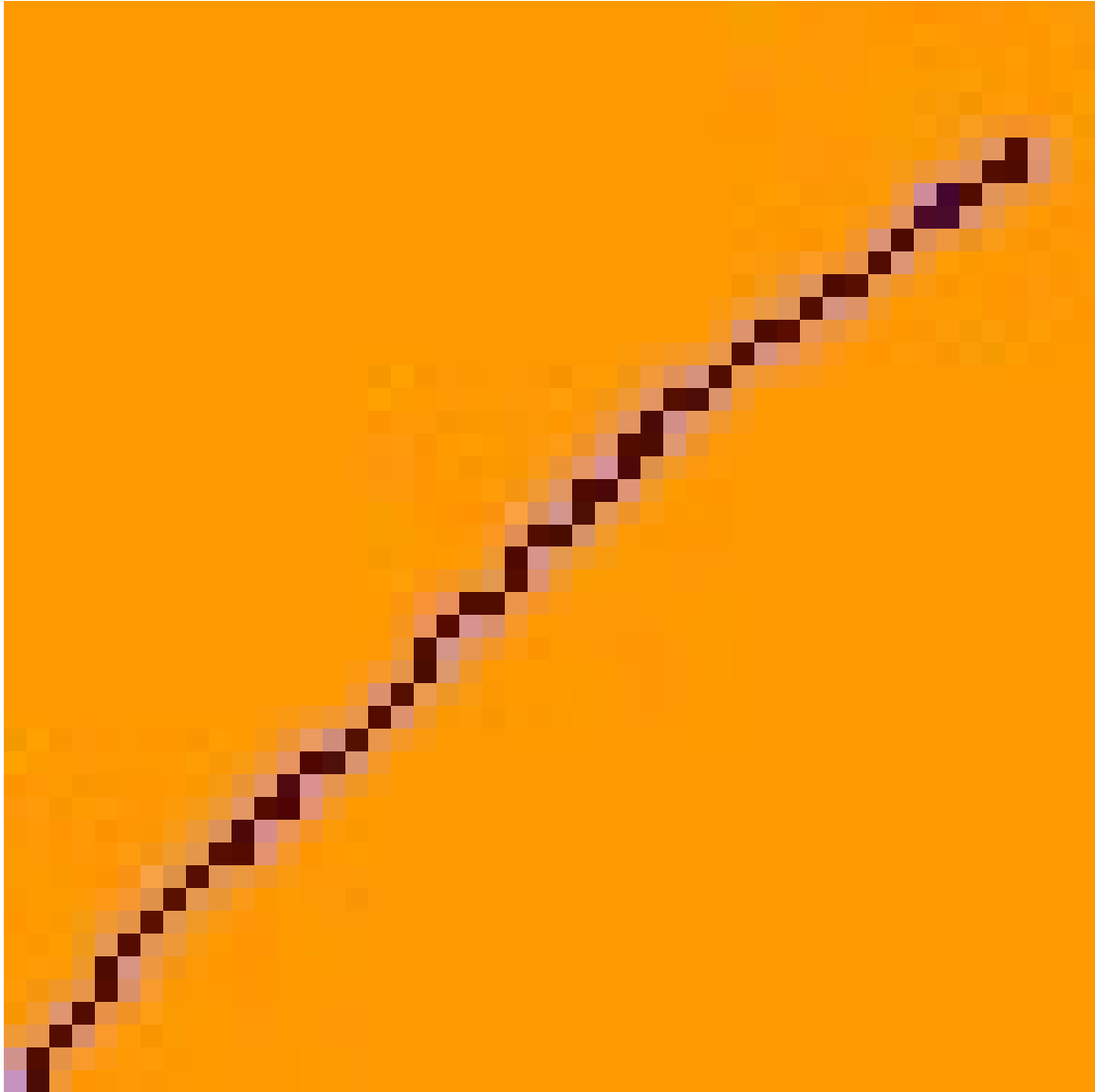


Figure 4: Distribution of DTA-JEPA’s online adaptation statistics across all conditions: surprise, allocated gradient steps, learning-rate multiplier, number of updates and retrieved memories. The allocation is strongly non-uniform, which is the mechanism the method claims.

6 Discussion and Conclusion

6.1 What the results say

6.2 Limitations

Protocol-level, not benchmark-level, replication. Our environments are a numpy re-implementation. The shift families, goal sampling, planner, budget and success criteria match AdaJEPA, but the physics, the observation statistics and the model architectures differ from PushT/PointMaze with DINO-WM or ResNet encoders trained on tens of thousands of expert trajectories. Absolute success rates are therefore not comparable across the two papers; the *ordering and the relative gains*, measured under identical conditions, are what our experiments establish.

Scale. Models are $\sim 1.3\text{M}$ parameters, trained on a few hundred trajectories per environment family on a laptop CPU/MPS. This is deliberate—it makes the study fully reproducible—but it also means the regime is data-poor, which is precisely where test-time adaptation is strongest. Whether the same allocation policy helps a 100M-parameter world model trained on tens of thousands of trajectories is an open question; our propositions suggest the mechanism should, but the operating point could shift.

Hand-set allocation shape. The mapping from relative surprise to steps, learning rate and refinement depth is a sigmoid with three hand-set ranges. It is not learned. A learned controller (e.g. bandit or meta-learned allocation) is an obvious next step, and the anchor ball gives the safety guarantee that would make exploration of larger budgets admissible.

Reward-free scope. We never use reward labels, which is a strength for generality and a limit for tasks where task-specific value information would dominate.

6.3 Conclusion

Adaptive latent world models need three separations that a single fixed test-time gradient step cannot provide: a separation between the slow dynamics that should not move and the fast computation that should, a separation between environments where the model is already right and environments where it is wrong, and a separation between the current episode and the experience accumulated so far. DTA-JEPA implements those separations with a dual-timescale recursive predictor, an uncertainty-allocated adaptation controller, a persistent dual memory with a safety layer, and reports gains over a faithful AdaJEPA re-implementation across all four of its shift families—largest where the model is most surprised, smallest where it is already adequate—while ablations attribute the improvement to allocation, memory, safety and recursive depth respectively.

7 Implementation Details

7.1 Architecture summary

Component	Specification	Online?
Encoder trunk	4 conv stages (32/64/96/128), stride 2, GroupNorm, GELU	no (slow)
Encoder proprio	MLP(4→64→64)	yes (slow head)
Encoder head	Linear(2048+64 → 256) GELU Linear(256 → d), LayerNorm(d)	yes (fast)
S-module	Linear($2d \rightarrow d$) + 2 pre-norm transformer blocks, $K = 3$ tokens	no (slow)
R-module	1 pre-norm transformer block over 2 tokens + LoRA($r=4$)	yes (fast)
Heads	mean Linear($d \rightarrow d$), log-variance Linear($d \rightarrow 1$)	yes (fast)
Action encoder	MLP(2 → 64 → d)	no
Inverse dynamics	MLP($2d \rightarrow 128 \rightarrow 2$)	no

$d = 128$ unless stated otherwise. Total 1.29M parameters, of which 594.7k are in the fast set (LoRA, encoder head, propriojection, R-module heads).

7.2 A note on LoRA inside `nn.MultiheadAttention`

Adding a LoRA adapter to a linear layer that is used as `out_proj` of PyTorch’s `nn.MultiheadAttention` fails: the attention implementation reads `.weight` and `.bias` attributes directly. Our adapter therefore exposes the *composed* weight $W + \frac{\alpha}{r}BA$ as a property, so gradients still flow to A and B while the existing attribute contract is preserved. This is a small but load-bearing implementation detail for anyone reproducing recursive-refinement predictors with LoRA.

7.3 Collapse in the two-term recipe and how we fixed it

Training a from-scratch JEPA with LeWM’s two-term objective collapsed in our setting: the encoder head has an unconstrained scale, so the cheapest way to reduce the prediction loss is to shrink every latent towards zero. We observed latents with per-dimension standard deviation ~ 0.004 that stayed there for hundreds of steps, because the isotropic-Gaussian regulariser’s restoring force is proportional to the latent scale and therefore vanishes exactly in the collapsed regime. Two fixes were required, and we report them because both matter in this regime:

1. a layer normalisation on the encoder output, which bounds the per-sample scale and removes the degenerate solution entirely;
2. a VICReg-style variance hinge $\frac{1}{d} \sum_i \text{relu}(1 - \text{std}_i(z))$ added to the regulariser, which supplies a scale-free gradient in the collapse regime.

With both in place the latent standard deviation rises from 0.004 to 0.97 within a few hundred steps and the prediction loss falls by more than an order of magnitude ($148 \rightarrow 3.3$ in our logs).

7.4 Hyper-parameters

Pretraining		
optimiser	AdamW, lr 5×10^{-4} , weight decay 10^{-4}	cosine decay to
batch size	48 windows (7 frames each)	
epochs	PushBlock 6, maze 8	
history / future / max refinement	3 / 4 / 4	
$\lambda_{\text{unc}}, \lambda_{\text{reg}}, \lambda_{\text{inv}}$	0.1 / 1.0 / 0.1	
Meta-training (FOMAML)		
inner lr / meta lr / iterations	5×10^{-4} / 10^{-4} / 300 (push), 200 (maze)	
tasks	leave-one-shape-out (4), layout groups (5)	
Online adaptation		
buffer / memory capacity / skill bank	5 (recent) / 256 / 8	
$U_{\text{min}}, U_{\text{max}}$	1, 3	
η multiplier range	$0.2\text{--}5 \times 5 \times 10^{-4}$	
$K_{\text{min}}, K_{\text{max}}$ (refinement depth)	1, 4	
$\lambda_{\text{ret}} / \epsilon_f$ / validation tolerance	0.5 / 0.8 / 0.05	
β (uncertainty) / γ (support)	0.5 / 0.5	
MC-dropout samples (epistemic)	4	
slow consolidation lr / EWC strength	10^{-6} / 10^3	
Planning		
planner / horizon / chunk / max replans	GD (Adam, lr 0.1, 40 steps) / 15 / 5 / 10	
goal gap / success tolerances	25 steps; block $\Delta p < 0.12$, $\Delta \theta < 0.4$ rad; maze < 0.9 cell	

7.5 Data generation and cost

Suites are generated by `python -m dtajepa.data` (about 8 minutes wall-clock on one M1 core, $\sim 100\text{k}$ rendered frames), pretraining by `python -m dtajepa.train` (about 30 minutes on Apple M1 MPS for both world models), meta-training by `python -m dtajepa.meta` (about 10 minutes), and evaluation by `python -m dtajepa.evaluate` followed by `python -m dtajepa.report`. Total disk footprint: 8.6 MB of compressed data, 5.2 MB per checkpoint.

8 Extended Analysis

8.1 Surprise as a detector of shift

The allocation policy assumes that surprise increases when the model is evaluated outside its training distribution. We can test the assumption directly, without any planner: for each suite, encode held-out episodes in distribution and under shift, compute the one-step Mahalanobis residual in latent space, and compare the distributions. The model is trained the same way in both cases; only the evaluation stream changes. This is the reward-free analogue of LeWM’s ”surprise evaluation”, and it is a pre-condition for using surprise as a control signal.

8.2 Allocation is non-uniform

If the controller were useless, allocated steps and refinement depth would be constant across episodes. Instead, the statistics collected during evaluation (Figure ??) show a spread of allocated steps, learning-rate multipliers and refinement depths, with the mean shifted upwards on shifted suites. The controller’s behaviour is additionally logged per replan, so the relationship between relative surprise and allocated compute can be read off directly from `results/raw.jsonl`.

8.3 Relation to AdaJEPA’s hand-tuned adaptation

AdaJEPA’s own ablations show that for out-of-distribution settings, ”stronger adaptation—via a larger learning rate, more optimization steps, or a larger recent replay buffer—is often beneficial when latency permits”, while in distribution a single step at the training learning rate is the practical default. That is exactly a two-point policy. DTA-JEPA can be read as learning the *shape* of that policy from a signal the agent already computes for free—the residual of its own prediction—instead of selecting one of two points by hand per environment family, and additionally spending recursive depth, which AdaJEPA’s architecture does not expose.

8.4 Failure modes we observed

- **Symmetric shapes are easier than they look.** The square and I blocks have $\pi/2$ rotational symmetry, so the orientation tolerance is met on a larger fraction of the pose space. Success rates on those shapes should not be read as geometric generalisation.
- **Colour shifts that remove the agent–distractor distinction are hardest.** Rendering the static reference marker in the same red as the moving block removes the visual cue that separates ”what moves” from ”what does not”, and no amount of predictor adaptation recovers it, consistent with AdaJEPA’s finding that red-anchor and red-block shifts are where their gains are smallest.
- **Replay-buffer entropy.** Because we execute five actions per replan, the recent-5 buffer covers one replanning step of experience, and its entropy is low. Memory-based retrieval is therefore most useful late in an episode or across episodes, which is what the skill bank targets.

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